Second Language Acquisition

Prelude: Babies are overrated.



When we talk about *sensitive periods* for L2, and say that infants/toddlers/children are “better” at learning language than adults…

…let’s not get carried away.

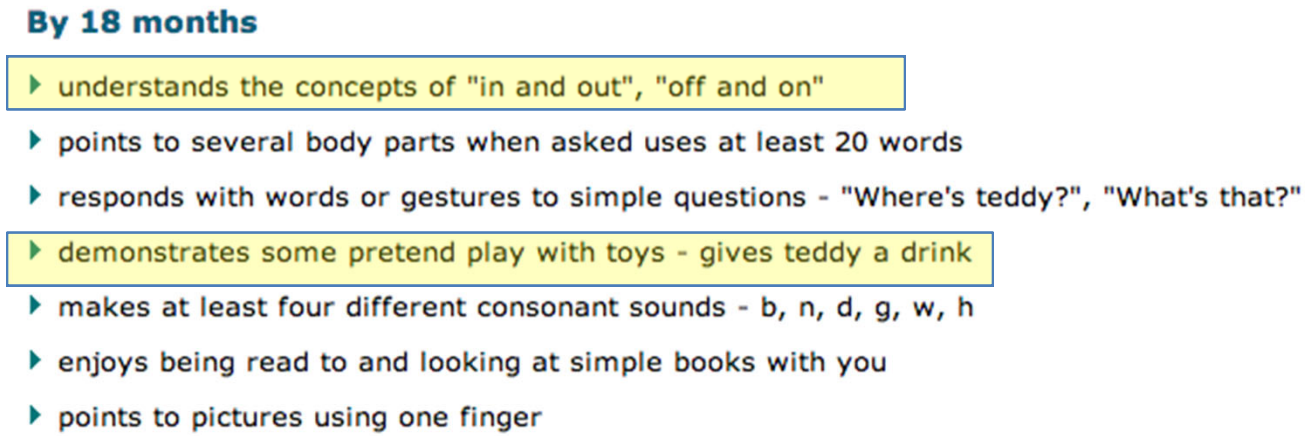
Imagine if you only knew about 20 words after 18 months of language instruction!

By 18 months

- understands the concepts of "in and out", "off and on"
- points to several body parts when asked uses at least 20 words
- responds with words or gestures to simple questions - "Where's teddy?", "What's that?"
- demonstrates some pretend play with toys - gives teddy a drink
- makes at least four different consonant sounds - b, n, d, g, w, h
- enjoys being read to and looking at simple books with you
- points to pictures using one finger

Adults initially learn more quickly than children. This is due largely to *transfer* from L1 and better general cognitive skills.

Where adults do worse is *ultimate attainment,* that is, rarely becoming native-like. This may also be due in part to transfer.



Consider that some of the “language milestones” here are actually general conceptual abilities, like the ability to deal with spatial relations between objects, or event representations.

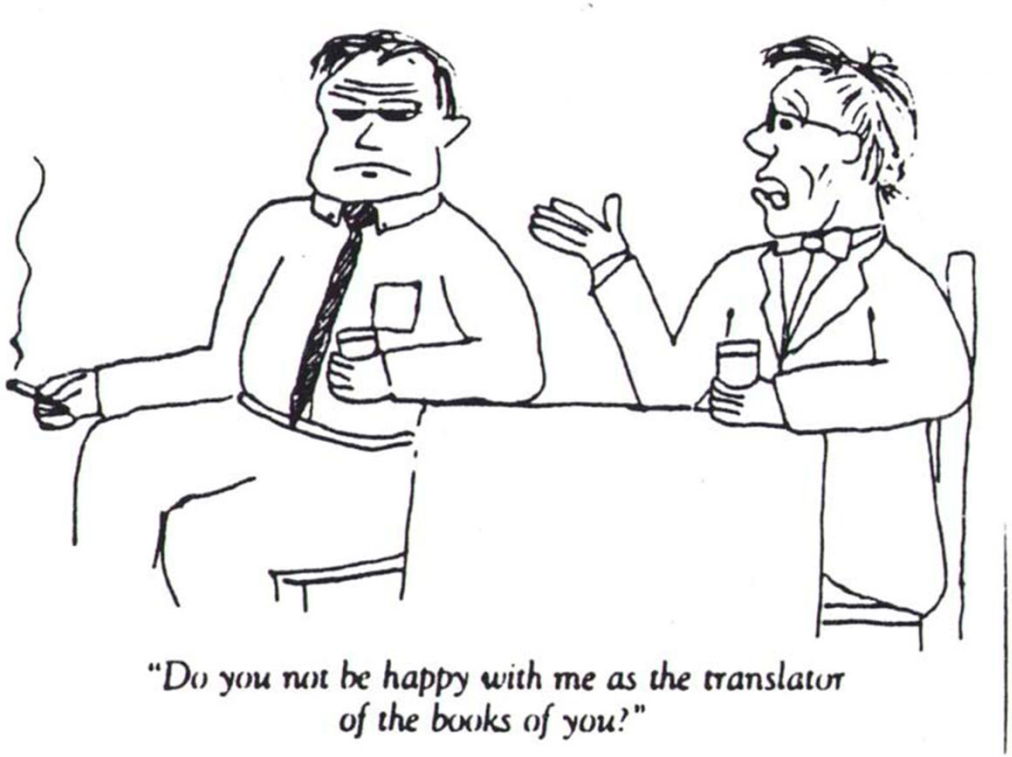
As later learners of a second language, we have a lot more of that stuff in place, and can focus on learning how they are represented in the target language.

Of course, we also already know another language, which has an impact on how the second language is learned…

Transfer

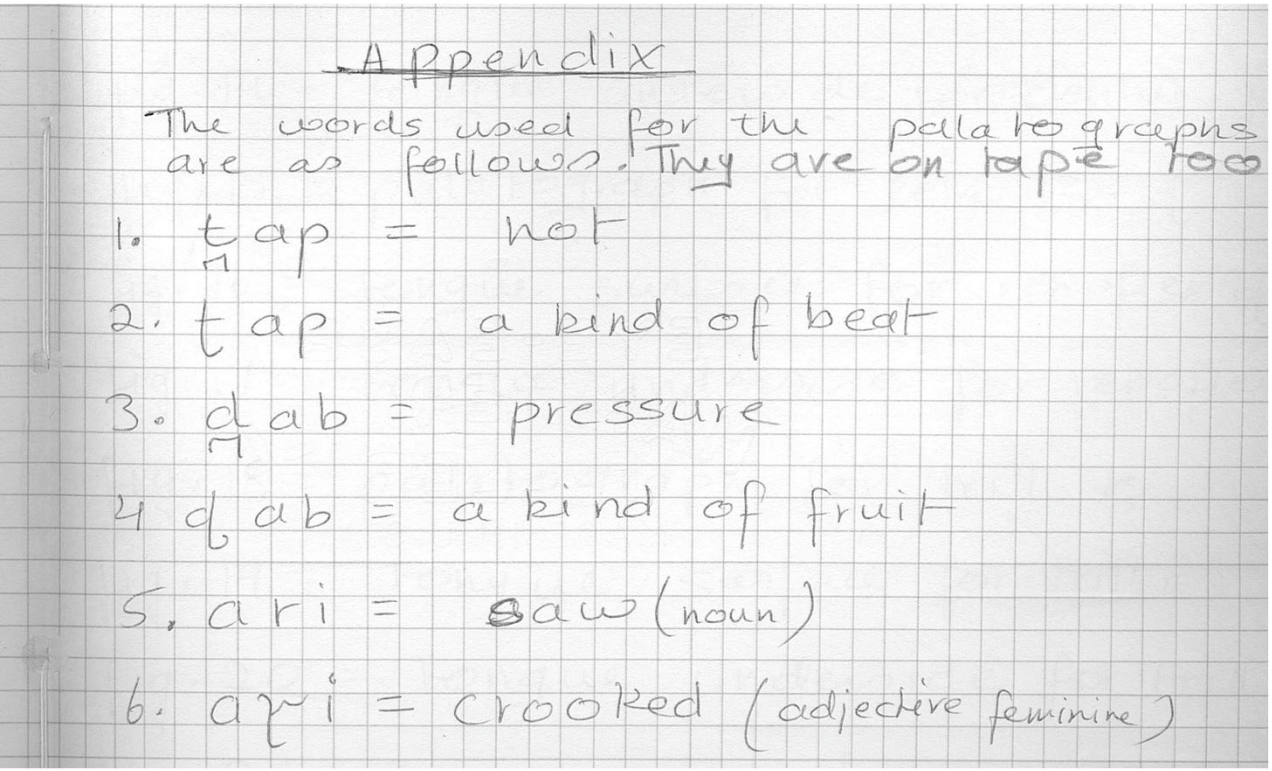
Transfer: The use of native language (L1) processes and representations for a second language (L2).

Transfer can have both *positive* and *negative* impacts on second language learning and performance:



"Do you not be happy with me as the translator of the books of you?"

Transfer and phonetic/phonological learning



We've talked a lot about categorical perception and the notion of *acquired similarity* for sounds in the same category.

Here are some minimal pairs in Hindi.

Native English speakers tend to use our "alveolar" category to interpret both dental and retroflex consonants.

So in this case, transfer makes learning the L2 contrast more difficult.

Listen to sounds from many, many languages here: http://archive.phonetics.ucla.edu/

Transfer and grammar learning

demonstrative pronoun + nominal classifier + noun

一个人

yī gè rén

Classifier	Use
个 *gè*	General classifier
把 *bǎ*	Used with objects that have handle
本 *běn*	Used books and similar objects
块 *kuài*	Used with objects that are divided into pieces
辆 *liàng*	Used with wheeled vehicles
台 *tái*	Used with devices and gadgets
些 *xiē*	Used to indicate indeterminate amount or plural of pronouns and demonstrative adjectives
张 *zhāng*	used with flat objects
支 *zhī*	Used with long thin objects

Classifiers are a somewhat unusual grammatical feature cross-linguistically.

Native English speakers learning Chinese (for example) have difficulty learning these.

In English, when we want to say “a chair,” “that dog,” “the car,” or “one book,” we don’t worry too much about noun semantics when picking a determiner or pronoun.

But in Chinese, different *classifiers* are required by different nouns.

One way to think about this is that English speakers have learned to *ignore* the different cues for classifiers as part of learning English grammar.

Gender of Nouns • Genre des Noms

Masculine		Common Endings Used With Masculine Nouns:	
le cheval[2]	the horse	-age	le fromage the cheese
le chien	the dog	-r	le professeur[3] the teacher
le livre	the book	-t	le chat the cat
le bruit	the noise	-isme	le capitalisme capitalism
Feminine		**Common Endings Used With Feminine Nouns:**	
la colombe	the dove	-ie	la boulangerie the bakery
la chemise	the shirt	-ion	la nation the nation
la maison	the house	-ite/-ité	la fraternité brotherhood
la liberté	liberty	-nce	la balance the scales
		-nne -mme -lle	la fille the girl l'indienne the Indian

Another example where transfer causes problems is with grammatical gender.

Grammatical gender is different from social/cultural gender.

There's nothing "male" about a *book* or "female" about *liberty* (although: hold that thought for when we discuss "language and thought" later in the semester).

But grammatical gender has implications for agreement. Pronouns, adjectives, some forms of verb, etc. all need to be coordinated with noun gender.

French is famous for having synonyms with different gender:

auto/voiture
rivière/fleuve
vélo/bicyclette

Transfer is not all bad.
Vocabulary learning: Cognates and loan words



auto



rivière



bicyclette

Vocabulary learning in some L1-L2 situations is aided by the similarity of words across languages.

Sometimes this is due to shared etymology, sometimes it's due to more recent "borrowing."



コロッケ (korokke)



モノレール (monoreru)

Not all similar-sounding words are cognates.



"Arnold esta embarazado."

The bias to interpret L2 words as related to similar-sounding L1 words can be very strong.

Sometimes the result is…embarrassing.

Transfer in the lab:
Do people use cognates in "second language" learning?

DeGroot & Keijzer (2000, Language learning)

- paired associate learning of made-up words
- varying in their similarity to their Dutch associates
- also varying in "concreteness."

For example:

gitaar – gettar ("guitar," concrete, cognate)
tijger - duunze ("tiger," concrete, non-cognate)
domein – domaan ("domain," abstract, cognate)
herstel – groop ("repair," abstract, non-cognate)

Participants were simply presented with pairs of Dutch words and their non-word paired associates, and asked to learn these pairs.

They were then tested a week later on retention.

Transfer in the lab:
Do people use cognates in "second language" learning?

Table 2

Recall scores and response times for all concreteness-by-cognate-status-by-session conditions (Experiment 1a)

	Recall scores (%)							
	Session 1		Session 2		Session 3		Session 4	
	C	NC	C	NC	C	NC	C	NC
Concrete	69.7	43.7	91.7	79.7	93.4	92.0	78.4	69.0
Abstract	42.7	19.0	74.7	53.4	86.0	72.7	68.4	36.0
	Response times (ms)							
	Session 1		Session 2		Session 3		Session 4	
	C	NC	C	NC	C	NC	C	NC
Concrete	1,880	2,556	1,318	1,823	1,080	1,643	1,833	2,597
Abstract	2,529	3,641	2,130	3,148	1,663	2,183	2,333	3,412

Note. C = Cognates. NC = Noncognates.

Cognates are learned more rapidly, and also remembered more easily.

So, this is evidence that cognate status can contribute to the very earliest stages of word learning in a second language.

(Session 4 is after one week with no training)

(We're not going to talk about concreteness effects in any detail, but if you're curious about why they have such a big effect, look up "context availability")

Is this a good experiment?

Concrete cognates
broer-breur
brood-brodde
gitaar-gettar
jongen-jange
lepel-lippel
lichaam-lechem
naald-nadel
paraplu-parredu
rivier-riffer
schaar-skare
stier-stire
trein-trinen
varken-vark
viool-vialo
vrouw-vrauk

If de Groot and Keijzer had only tested “receptive” vocabulary – how easily participants could produce the *Dutch* word given its associate – this would not be very impressive.

Consider that you might have guessed the meaning of *gitaar, tijger* and *domein* without any Dutch training.

But the study also involved a test of “expressive” vocabulary (what they call “productive” testing).

Table 3

Recall scores (in percentages) immediately following learning (Session 3) and at retest (Session 4) for abstract vs. concrete words and for cognates vs. noncognates (Experiment 1a: productive testing; Experiment 1b: receptive testing)

	Productive testing				
	Concrete	Abstract		Cognates	Noncognates
Session 3	92.7	79.3	Session 3	89.7	82.3
Session 4	73.7	52.2	Session 4	73.3	52.5
Effect	19.0	27.1	Effect	16.4	29.8
	Receptive testing				
	Concrete	Abstract		Cognates	Noncognates
Session 3	97.7	91.5	Session 3	98.5	90.7
Session 4	88.9	64.9	Session 4	90.7	63.0
Effect	8.8	26.6	Effect	7.8	27.7

What does this table tell you about what participants actually learned in the study?

Research question:
Are similar L1 words always activated when processing L2?

There have been many, many studies of this.

One nice example comes from a “visual world” experiment by Marian & Spivey (2003, Bilingualism: Language and Cognition)

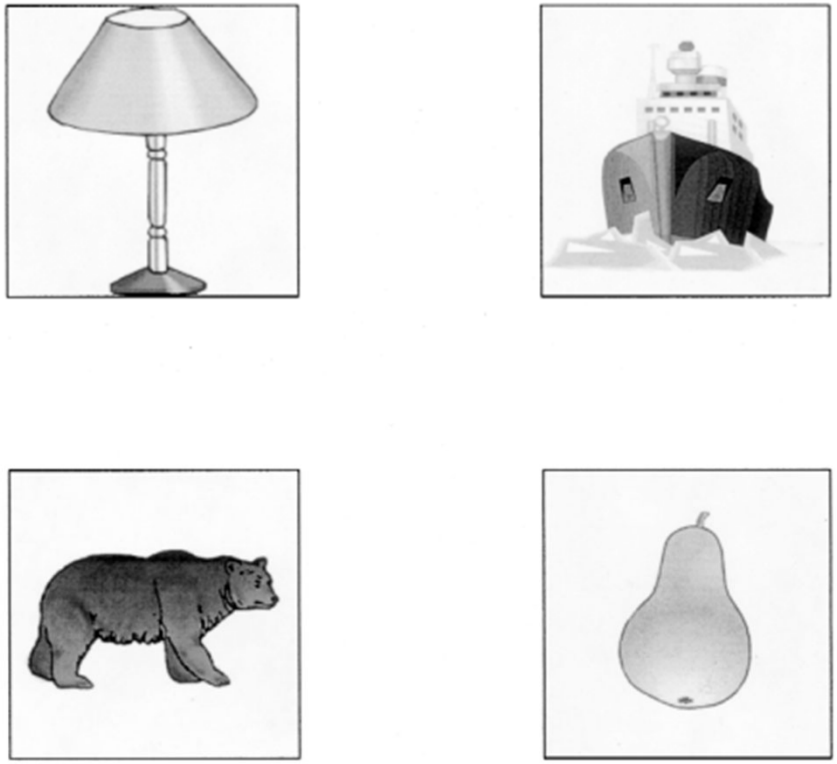


Fig. 3. A sample display. Each display contains a target (e.g., the "pear"), a competitor (e.g., the "bear"), and two filler items ("lamp" and "ship").

McMurray, B., Tanenhaus, M., Aslin, R. and Spivey, M. (2003). Probabilistic constraint satisfaction at the lexical/phonetic interface: Evidence for gradient effects of within-category VOT on lexical access. *Journal of Psycholinguistic Research, 32*, 77-97.

http://www2.psychology.uiowa.edu/faculty/mcmurray/publications/mcmurray-jpr.pdf

Remember that you've seen this kind of experiment before, when we talked about categorical perception.

Eye movements can tell us about competition between possible responses participants are considering.

In the McMurray et al. study, the amount of competition between BEAR and PEAR seemed to depend on distance from the category boundary.

Remember?

The “cohort effect.”

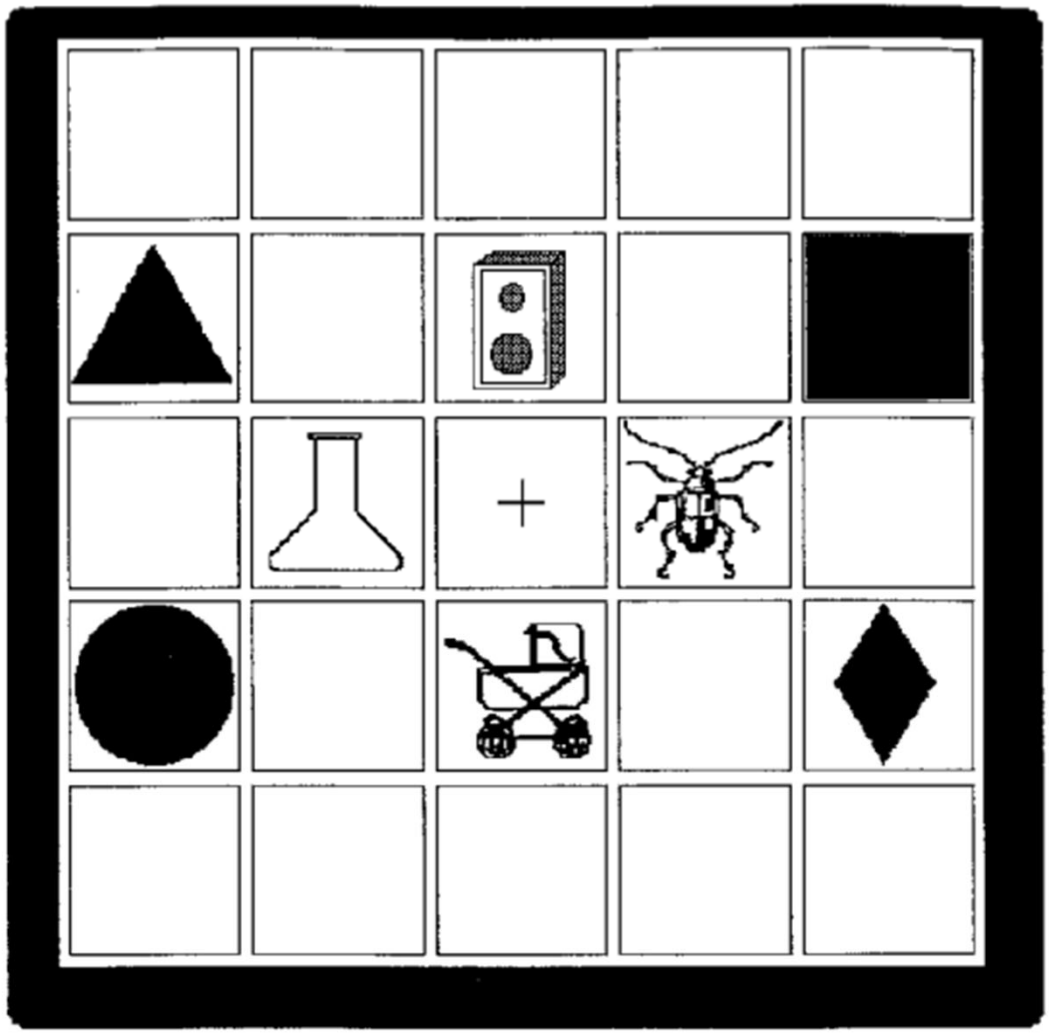


FIG. 3. An example of a stimulus display presented to participants.

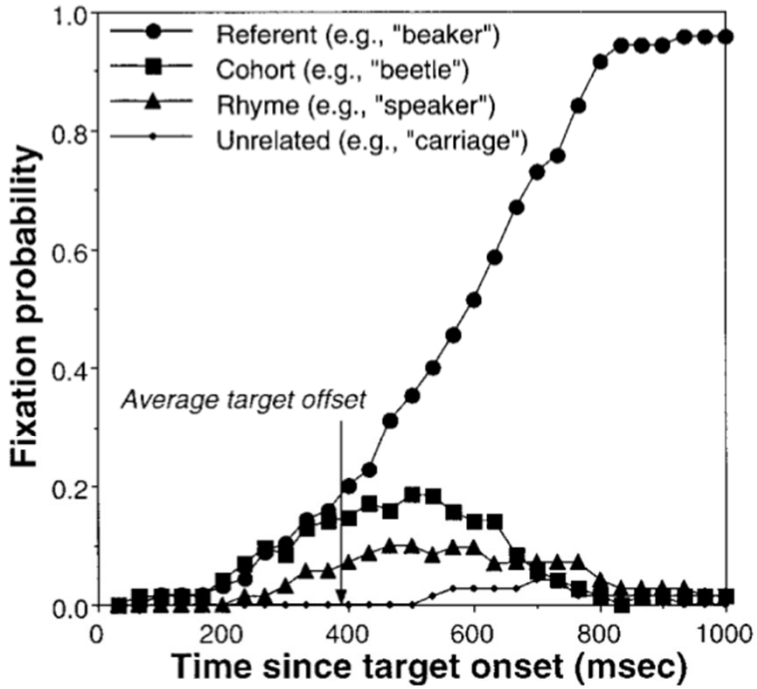


FIG. 4. Probability of fixating each item type over time in the full competitor condition in Experiment 1. The data are averaged over all stimulus sets given in Table 1; the words given in the figure are examples of one set.

Allopenna et al. (1994, JML)

Marian & Spivey examined the cohort effect with "cross-language competitors"

Russian Target			English Competitor	Russian Competitor		
Спички	[spichki]	(Matches)	Spear	Спицы	[spitsi]	(Knitting needles)
Бусы	[boosy]	(Beaded necklace)	Book	Бубен	[buben]	(Tambourine)
Черепаха	[cherepaha]	(Turtle)	Chair	Червяк	[chervyak]	(Worm)
Марка	[marka]	(Stamp)	Marker	Марля	[marlya]	(Cheese cloth)
Баран	[baran]	(Ram)	Barbed Wire	Бархат	[barhat]	(Velvet)
Платье	[platye]	(Dress)	Plum	Плащ	[plashch]	(Raincoat)
Гайка	[gaika]	(Nut)	Gun	Галстук	[galstuk]	(Necktie)
Карта	[karta]	(Map)	Car	Картошка	[kartoshka]	(Potato)
Лак для ногтей	[Lak dlea nogtei]	(Nailpolish)	Lock	Лопата	[lapata]	(Shovel)

English Target	English Competitor	Russian Competitor		
Speaker	Spear	Спички	(spichki)	Matches
Boot	Book	Бубен	(buben)	Tambourine
Shark	Shovel	Шарик	(sharik)	Balloon
Chair	Chess set	Черепаха	(cherepaha)	Turtle
Marker	Marbles	Марка	(marka)	Stamp
Barbed wire	Bark	Бархат	(barhat)	Velvet
Plug	Plum	Платье	(platye)	Dress
Gun	Gum	Гайка	(gaika)	Nut
Lock	Lobster	Лак для ногтей	(Lak dlea nogtei)	Nailpolish

Hypothesis: If L1 words are always activated, native Russian speakers should look at, e.g., a book of matches when the target is "speaker" in English.

Competition during L2 spoken word recognition

Table 2. *Percent of trials in which bilingual participants made eye movements to the competitor items and their corresponding fillers in between-language competition, within-language competition, and simultaneous competition trials in Experiment 1.*

Display	Fixations of between-language competitor	Fixations of within-language competitor
Between-language competitor present	18%	n/a
Within-language competitor present	n/a	18%
Both competitors present	13%	19%
No competitor present; fixations of control filler object	7%	

When only one competitor was present, participants made eye movements to both within- and between-language competitors to the same extent.

When both within- and between-language competitors were present, participants looked at the within-language competitor more, but still looked at the between-language competitor a lot (i.e., twice as often as a control condition).

Competition during **L1** spoken word recognition

Table 4. *Percent of trials in which bilingual participants made eye movements to the competitor items and their corresponding fillers in between-language competition, within-language competition, and simultaneous competition trials in Experiment 2.*

Display	Fixations of between-language competitor	Fixations of within-language competitor
Between-language competitor present	8%	n/a
Within-language competitor present	n/a	14%
Both competitors present	8%	13%
No competitor present; fixations of control filler object	5%	

In contrast, English words do not seem to be activated when (native Russian speakers) do this experiment in Russian.

(The small numerical difference between the between-language and control conditions is not statistically significant.)

A model explaining L1-L2 interactions in terms of spreading activation:
The revised hierarchical model of Kroll & Stewart

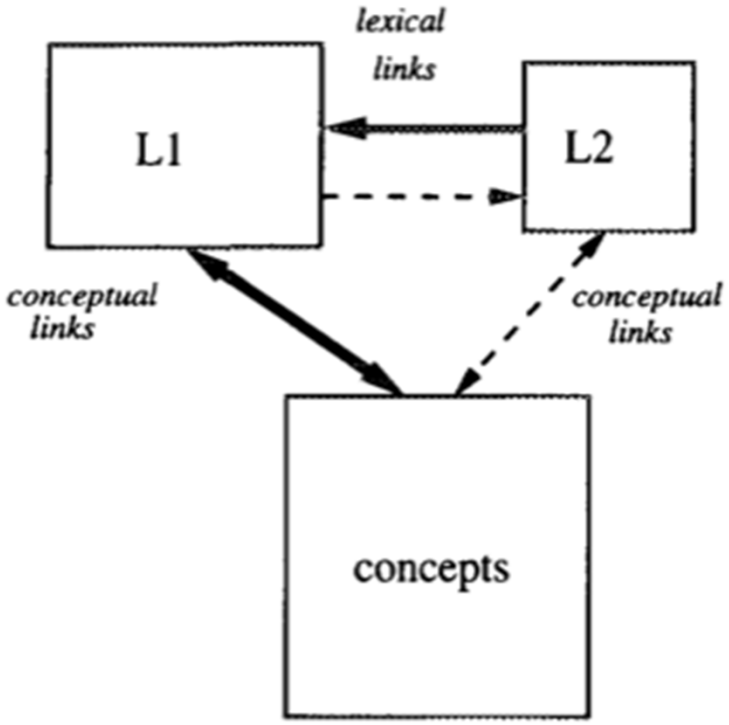


FIG. 3. Revised hierarchical model of lexical and conceptual representation in bilingual memory.

The L1 lexicon is assumed to be:

- larger
- more strongly connected to relevant concepts

The L2 lexicon is assumed to have strong *forward* connections to the L1 lexicon.

That is, activation can spread more readily from an L2 word to an L1 word than the other way.

(This occurs because this direction is emphasized in learning.)

A model explaining L1-L2 interactions in terms of spreading activation:
The Revised Hierarchical Model of Kroll & Stewart (1990; 1994)

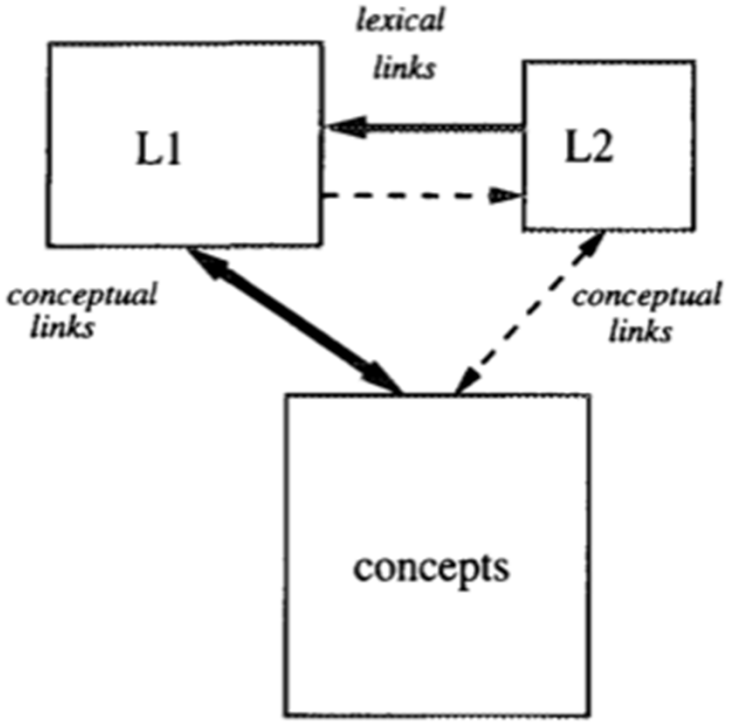


FIG. 3. Revised hierarchical model of lexical and conceptual representation in bilingual memory.

Consider what happens in this model when performing the Marian & Spivey task in L2:

- the concept for “matches” partially activates the Russian word for “matches”

- this activity feeds back to the conceptual activation for “matches”

- conceptual activation for “matches” leads to eye-movements toward the L1 competitor.

When the task is done in L1, we might imagine that the conceptual links with L2 are too weak to create this kind of interference reliably.

Competition, interference and bilingualism:

Note that this model makes a counter-intuitive prediction even for “balanced bilinguals,” i.e., that they do not look like “two monolinguals in one,” but rather that competitive dynamics and inhibition will be engaged all the time b/c the conceptual links are similarly strong for both languages.

This has lead to a lot of work on bilinguals’ advantages in non-linguistic cognitive control tasks…

So far we've considered a particular sequence of second language learning.

	learn L2
begin learning L1	*continue learning/using L1*



time

But what happens when you stop using L1 after L2 is introduced?

	learn L2
begin learning L1	~~*continue learning/using L1*~~

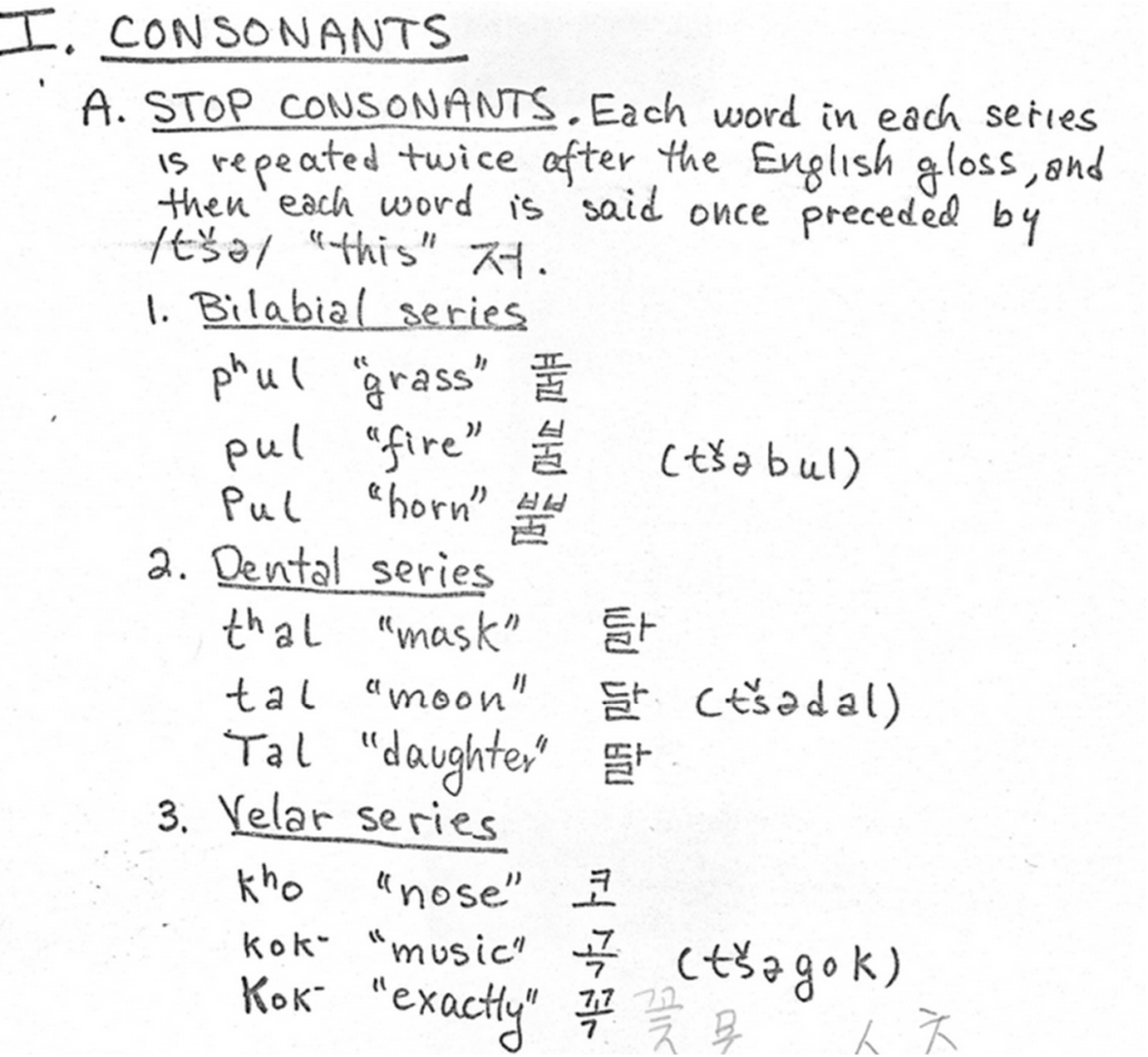
Brain Imaging of Language Plasticity in Adopted Adults: Can a Second Language Replace the First?

C. Pallier[1,2], S. Dehaene[1], J.-B. Poline[1], D. LeBihan[1], A.-M. Argenti[2], E. Dupoux[2] and J. Mehler[2]

[1]INSERM U562, Service Hospitalier Fredrik Joliot, CEA/DSV/DRM, & IFR49, 4 place du Général Leclerc, Orsay and [2]Laboratoire de Sciences Cognitives et Psycholinguistique, EHESS-CNRS, 54 bd Raspail, Paris, France

Adults (ages 20-32) who had been adopted into French homes from Korean orphanages between the ages of 3 and 8 were tested to see if they had any residual knowledge of Korean…

Korean three-way stop contrast.



I. CONSONANTS

A. STOP CONSONANTS. Each word in each series is repeated twice after the English gloss, and then each word is said once preceded by /tɕʌ/ "this" 저.

1. Bilabial series

pʰul	"grass"	풀	
pul	"fire"	불	(tɕʌbul)
Pul	"horn"	뿔	

2. Dental series

tʰal	"mask"	탈	
tal	"moon"	달	(tɕʌdal)
Tal	"daughter"	딸	

3. Velar series

kʰo	"nose"	코	
kok-	"music"	곡	(tɕʌgok)
Kok-	"exactly"	꼭	

(more good stuff from the UCLA Phonetics Archive)

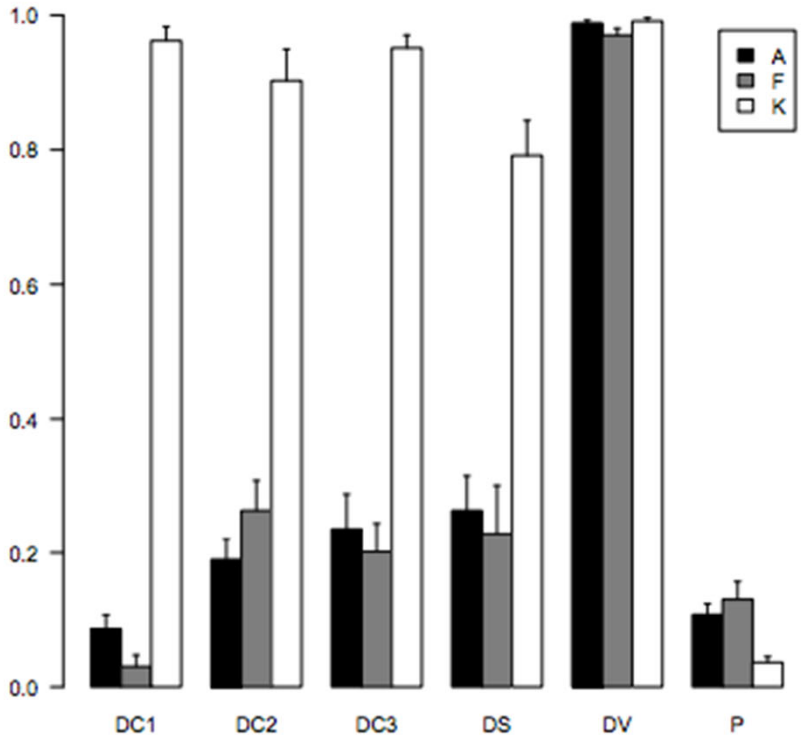
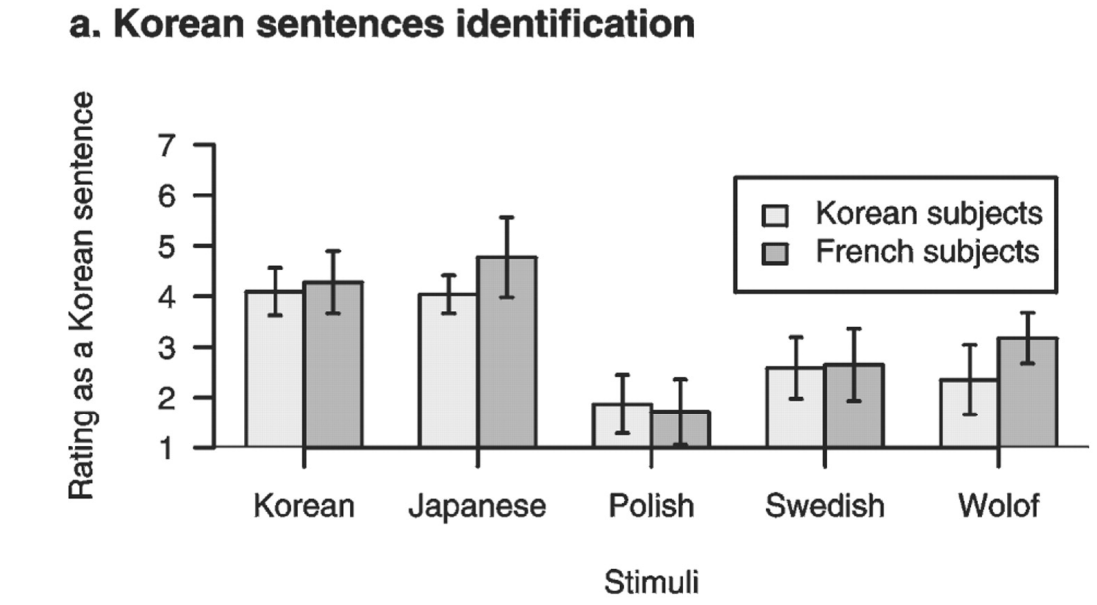
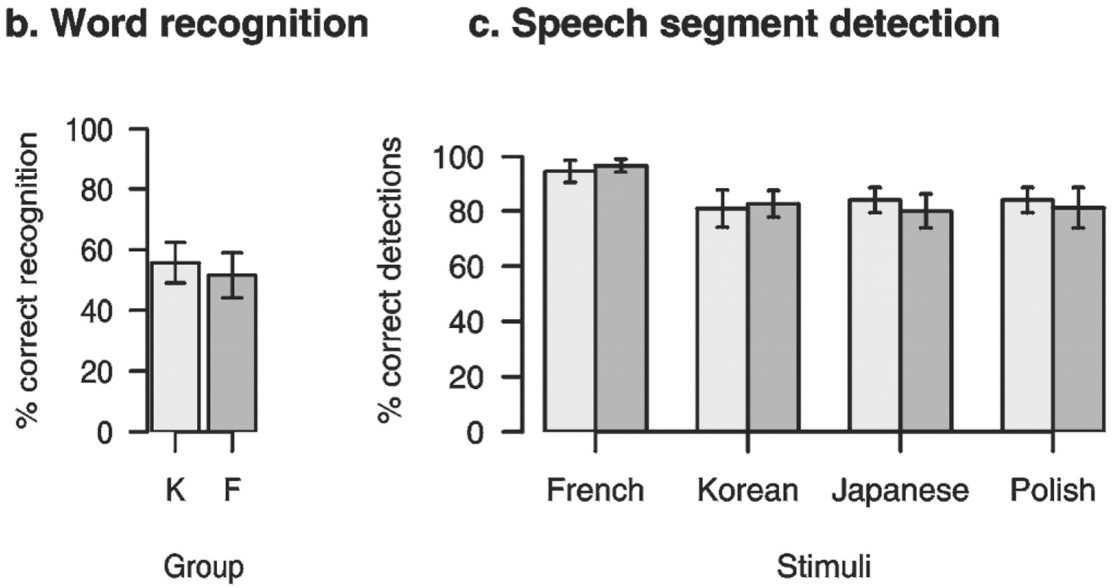


Figure 3. Results of the Phonetic Discrimination task for three groups of subjects (A= adoptees, F= French native speakers, K= Korean native speakers) expressed as percentage of different responses per category of contrast. (DC1=plain-tense contrast for p and k (11); DC2=plain-aspirated p and k (12); DC3=tense-aspirated p and k (12); DS=plain-tense s (4); DV=vowel contrast (48) and P=Same(48)).

Ventureyra et al. found that Korean adoptees in France were no better than monolingual French speakers at these contrasts.





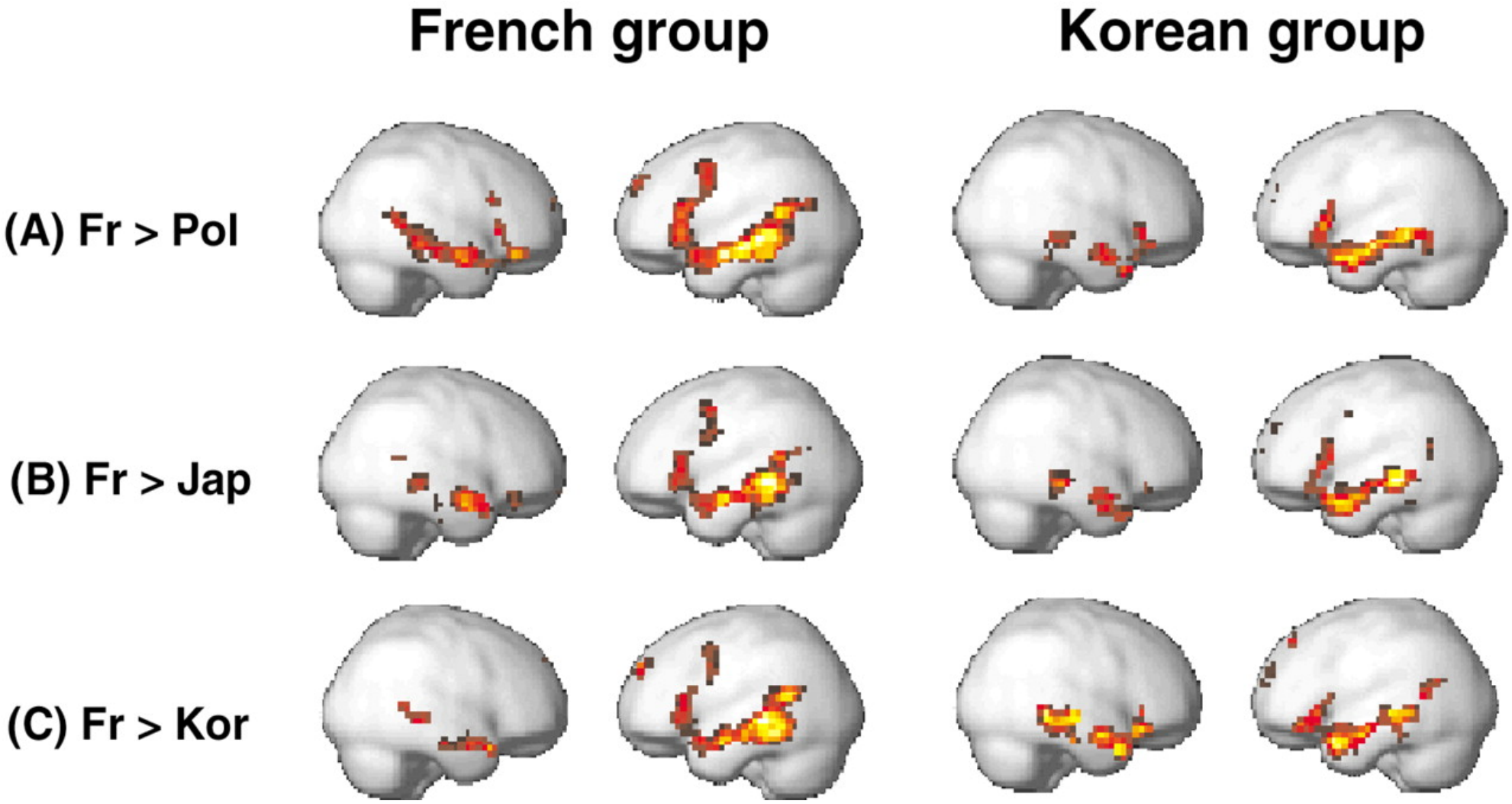
In a more extensive study, Pallier et al. found that adoptees were no better than monolingual francophones at:

a) Distinguishing Korean from a collection of largely unrelated languages.

b) Identifying the correct Korean translation of French words, or

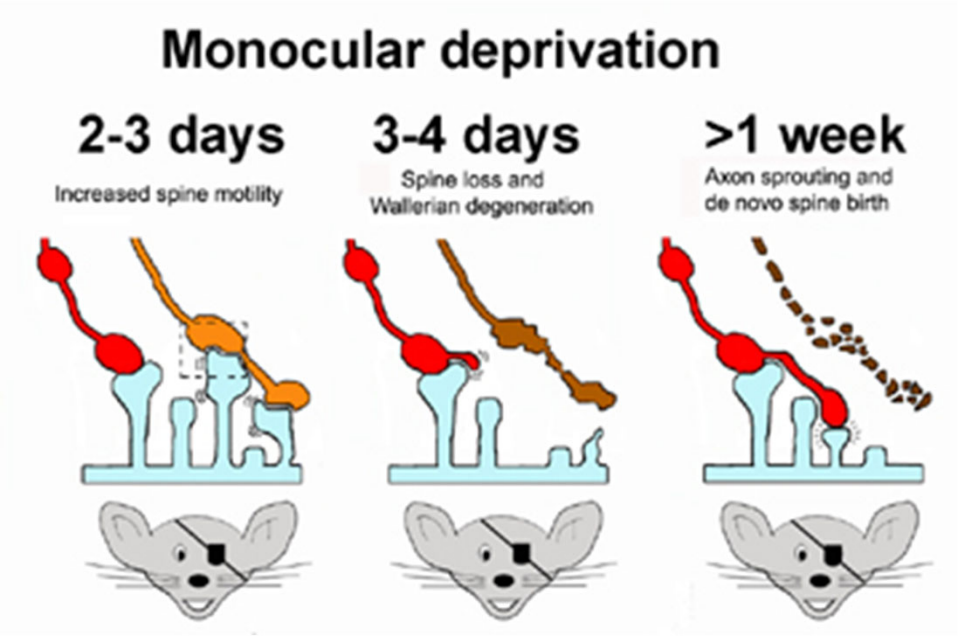
c) Determining whether a short excerpt was repeated from a recorded sentence they just heard.

These are not standard psycholinguistic tasks! Pallier et al. were looking for any remaining trace of Korean, and didn't find any.



Pallier et al. also found that listening to French activated the (now familiar?) classical language network, including superior temporal and inferior frontal regions.

Subtracting Korean from French might have revealed some difference between groups, if there were any residual processing ability for Korean left. But it didn't.



We can also think of L2-L1 interactions in terms of competitive dynamics.

Imagine that L1 is represented as a set of connections within/between different brain regions.

If those connections are *never activated again*, they could be “taken over” by L2.

Brain Imaging of Language Plasticity in Adopted Adults: Can a Second Language Replace the First?

Apparently YES!

Are traces of L1 left behind that can be “salvaged,” though?...

PSYCHOLOGICAL SCIENCE

Research Article

OVERHEARING A LANGUAGE DURING CHILDHOOD

Terry Kit-fong Au, Leah M. Knightly, Sun-Ah Jun, and Janet S. Oh

University of California, Los Angeles

Table 1. *Participants' performance on tests of Spanish abilities*

Measure	Native speakers	Childhood overhearers	Typical late L2 learners
Childhood slang			
Production	79_a (10)	17_b (3)	3_c (2)
Comprehension	88_a (4)	22_b (4)	3_c (10)
Voice onset time (ms)			
Word-initial voiceless stops	19.3_a (1.1)	19.6_a (2.5)	36.2_b (4.8)
Word-medial voiceless stops	22.4_a (1.2)	23.4_a (4.0)	31.2_a (3.4)
Percentage lenited			
Word-initial voiced stops	43_a (9)	24_a (8)	1_b (1)
Word-medial voiced stops	72_a (3)	63_a (8)	23_b (5)
Accent rating			
Voiceless stops	4.3_a (0.06)	3.5_b (0.12)	2.2_c (0.15)
Voiced stops	4.4_a (0.08)	3.3_b (0.15)	2.7_c (0.12)
Morphology production			
Gender agreement	93_a (4)	70_b (5)	68_b (6)
Number agreement	92_a (5)	88_a (4)	85_a (5)
Grammaticality judgment			
Accuracy	94_a (1)	64_b (2)	63_b (3)
Reaction time (ms)	$1{,}042_a$ (190)	$2{,}475_b$ (374)	$2{,}209_{a,b}$ (304)

Note. For the measures of childhood slang, morphology production, and accuracy of grammaticality judgment, the table indicates the mean percentage correct. Accent ratings were on a 5-point scale, with higher ratings indicating more nativelike pronunciation. Standard errors are given in parentheses. Within a row, means with the different subscripts were reliably different from each other according to Tukey's HSD test, two-tailed, $p < .05$. Numbers with the same subscript were not reliably different from each other.

More recent work from Terry Au's group (just across town) has examined this.

Undergraduate students who *overheard* Spanish frequently as children, but never learned to speak the language were studied.

All participants started classroom instruction of Spanish at around age 14, and were tested at the end of a Year 2 college Spanish course.

Taking the data from Korean adoptees together with the data from overhearers, it looks like:

If L1 is removed from the environment entirely, L2 can appear to *completely replace it.*

But this replacement is not necessarily complete. There may be some residual knowledge that can be "unmasked" with later learning.

Consider some of the potentially important differences between these populations, though:

Taking the data from Korean adoptees together with the data from overhearers, it looks like:

If L1 is removed from the environment entirely, L2 can appear to *completely replace it.*

But this replacement is not necessarily complete. There may be some residual knowledge that can be "unmasked" with later learning.

Consider some of the potentially important differences between these populations, though:

- Korean is rarely heard or encountered in France. Not so for Spanish in Los Angeles
- Differences between a "heritage language" spoken by other family members and a language not spoken by anyone you know
- Cognates
- Shared writing system

…etc.

This is a general issue with work on second language acquisition/bilingualism. Languages and learning situations differ in many, many ways that are hard (impossible?) to control experimentally….